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Inventor(s)	Parker; Charles Michael

Mounting system for seat frame structure

Abstract

Described are aircraft seat frame assemblies comprising a tube and a seat frame component mounting system. The seat frame component mounting system comprises a first tube mounting member and a second tube mounting member separable from the first tube mounting member. The tube is positionable between the first tube mounting member and the second tube mounting member. The seat frame component mounting system further comprises a fastener extending at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member. A clamping force is substantially evenly applied from the second tube mounting member and the first tube mounting member against an outer circumferential surface of the tube via the fastener.

Inventors:	Parker; Charles Michael (Irvine, CA)
Applicant:	Safran Seats USA LLC (Gainesville, TX)
Family ID:	1000008819277
Assignee:	Safran Seats USA LLC (Gainesville, TX)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3221743	12/1964	Thompson	600/233	A61G 13/10
4297922	12/1980	Higdon	81/57.2	E21B 19/164
4375300	12/1982	Long	297/445.1	B64D 11/0648
4489978	12/1983	Brennan	244/118.6	B64D 11/06
4617916	12/1985	LeVahn	600/231	A61B 90/50
4718719	12/1987	Brennan	296/68.1	B64D 11/0696
4783097	12/1987	Browning	D12/222	B60R 13/00
5038430	12/1990	Bly	5/430	A61G 7/0507
5224680	12/1992	Greenstein	600/230	A61B 90/50
5297890	12/1993	Commins	403/384	F16L 3/12
5553923	12/1995	Bilezikjian	297/452.2	B64D 11/0696
5888197	12/1998	Mulac	403/396	A61B 17/02
5897087	12/1998	Farley	248/316.2	A61B 90/50
5921629	12/1998	Koch	297/440.16	B64D 11/06
6033363	12/1999	Farley	600/234	A61B 90/57
6974276	12/2004	Kirchner	248/62	F16B 7/0486
8172840	12/2011	Murner	403/384	A61B 17/6466
8393680	12/2012	Zimmermann	297/248	B64D 11/0693
2003/0094838	12/2002	Williamson	297/232	B64D 11/0696
2003/0214153	12/2002	Kanie	296/193.07	B62D 25/2072
2008/0247818	12/2007	Oesch	403/389	F16B 2/065
2008/0312509	12/2007	Jacobson	600/230	A61B 90/57
2009/0088751	12/2008	Mullaney	606/54	A61B 17/6466
2012/0004659	12/2011	Miller	606/54	A61B 17/60
2017/0021933	12/2016	Pozzi	N/A	B64D 11/0648
2018/0244393	12/2017	Malligere	N/A	F16B 7/0433
2019/0002106	12/2018	Stachel	N/A	B64D 11/0648
2020/0010037	12/2019	Davenport	N/A	B60R 16/08
2021/0186562	12/2020	Mullaney	N/A	A61B 17/6425
2021/0190261	12/2020	Warner	N/A	F16M 11/16
2021/0330420	12/2020	Nowak	N/A	A61B 17/0206
2022/0349433	12/2021	Wang	N/A	F16B 2/10
2023/0098104	12/2022	Nowak	600/229	F16B 2/185
2023/0320715	12/2022	Martin	600/231	A61B 90/50
2024/0140607	12/2023	Madrid	N/A	B64D 11/0619
2024/0182172	12/2023	Omoragbon	N/A	B64D 11/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
647356	12/1949	GB	N/A
894214	12/1961	GB	N/A
07215220	12/1994	JP	N/A

Primary Examiner: Islam; Syed A

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

FIELD OF THE INVENTION

(1) The field of the invention relates to passenger seats, and, more particularly, to seat frame mounting assemblies for passenger seats.

BACKGROUND

(2) Passenger vehicles, such as aircraft, buses, trains, ships, and automobiles, often include one or more passenger seats in which passengers may be seated and otherwise use during travel. A passenger seat may include a seat frame assembly that connects to and supports structural tubes of the passenger seat. Traditionally, such seat frame assemblies have been a single-piece construction with either a clamping system adjacent to a portion of the seat frame assembly in which a structural tube is positioned or a bolt-through design through the structural tube. The clamping system often generates stress corrosion cracking and other mechanical failures in the structural tube, and the bolt-through design permits undesired movement between the seat frame assembly and the structural tubes.

SUMMARY

(3) The terms “invention,” “the invention,” “this invention” and “the present invention” used in this patent are intended to refer broadly to all of the subject matter of this patent and the patent claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below. Embodiments of the invention covered by this patent are defined by the claims below, not this summary. This summary is a high-level overview of various aspects of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings and each claim.

(4) According to certain embodiments of the present invention, a passenger seat comprises: a tube; and a seat frame component mounting system comprising: a first tube mounting member; a second tube mounting member separable from the first tube mounting member, wherein the tube is positionable between the first tube mounting member and the second tube mounting member; and a fastener extending at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; and wherein a clamping force is substantially evenly applied from the second tube mounting member and the first tube mounting member against an outer circumferential surface of the tube via the fastener.

(5) In some embodiments, the first tube mounting member may be integrally formed with a seat frame component.

(6) In certain embodiments, an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member may be in contact with at least a portion of the outer circumferential surface of the tube.

(7) The least one of the first tube mounting member or the second tube mounting member, in some embodiments, comprises at least one guide member.

(8) The other of the at least one of the first tube mounting member or the second tube mounting member, in certain embodiments, comprises at least one opening.

(9) In some embodiments, the second tube mounting member is positionable relative to the first tube mounting member so that the at least one guide member substantially aligns with the at least one opening.

(10) In certain embodiments, the at least one second fastener is coupleable to the first tube mounting member and extends at least partially through the at least one opening, the at least one guide member, and the first tube mounting member.

(11) In some embodiments, there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.

(12) According to certain embodiments of the present invention, a method (which may incorporate features of any of the preceding or subsequent examples) for assembling a passenger seat comprising a tube and a seat frame component mounting system comprises: positioning the tube between a first tube mounting member and a second tube mounting member, wherein the second tube mounting member is separable from the first tube mounting member; inserting a fastener at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; and tightening the fastener to apply a clamping force from the second tube mounting member and the first tube mounting member substantially evenly against an outer circumferential surface of the tube.

(13) In some embodiments, an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

(14) In certain embodiments, at least one of the first tube mounting member or the second tube mounting member comprises at least one guide member and at least one opening.

(15) The method, in some embodiments, further comprises aligning the first tube mounting member and the second tube mounting member by aligning the at least one guide member of the at least one of the first tube mounting member or the second tube mounting member with the at least one opening of at least one of the first tube mounting member or the second tube mounting member.

(16) The method, in certain embodiments, further comprises coupling at least one second fastener to the first tube mounting member, wherein the at least one second fastener extends at least partially through the at least one opening, the at least one guide member, and the first tube mounting member.

(17) In some embodiments, there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.

(18) According to certain embodiments of the present invention, a seat frame component mounting system comprises: a first tube mounting member; a second tube mounting member separable from the first tube mounting member and moveable relative to the first tube mounting member along a linear axis; and a fastener extending at least partially through the second tube mounting member and the first tube mounting member to couple the second tube mounting member to the first tube mounting member.

(19) In some embodiments, the first tube mounting member is integrally formed with a seat frame

component.

(20) The linear axis, in certain embodiments, extends substantially along a central axis extending through the fastener.

(21) In some embodiments, at least one of the first tube mounting member or the second tube mounting member comprises at least one guide member and at least one opening.

(22) The second tube mounting member, in certain embodiments, is positionable relative to the first tube mounting member so that the at least one guide member substantially aligns with the at least one opening.

(23) In some embodiments, there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a generalized passenger seat according to certain embodiments of the present invention.

(2) FIG. 2 is a partial side view of a seat frame component mounting system of the generalized passenger seat of FIG. 1 according to certain embodiments of the present disclosure.

(3) FIG. 3A is a side exploded view of the seat frame component mounting system of FIG. 2 with a seat tube according to certain embodiments of the present disclosure.

(4) FIG. 3B is a perspective exploded view of the seat frame component mounting system of FIG. 2 according to certain embodiments of the present disclosure.

(5) FIG. 4 is a side view of a first tube mounting member of the seat frame component mounting system of FIG. 2 according to certain embodiments of the present disclosure.

(6) FIG. 5 is a partial perspective view of the generalized passenger seat of FIG. 1 according to certain embodiments of the present disclosure.

(7) FIG. 6 shows a flowchart for an example method for assembling a passenger seat having a tube and a seat frame component mounting system according to certain embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

(9) The described embodiments of the invention provide seat frame component mounting systems for passenger seats. While the seat frame component mounting systems are discussed for use with aircraft seats, they are by no means so limited. Rather, embodiments of the seat frame component mounting systems may be used in passenger seats or other seats of any type or otherwise as desired.

(10) According to certain embodiments of the invention, e.g., as shown in FIG. 1, a generalized passenger seat **100** may include one or more seat frame component mounting systems **102** and one or more seat back assemblies **105** that may each include various components that provide support for a passenger in the generalized passenger seat **100**. Various components include, but are not limited to, one or more spreaders, arm rests, seat legs, etc. The generalized passenger seat **100** may be coupled to a seat track via a set of seat track fittings (e.g., a first seat track fitting and a second

seat track fitting).

(11) The generalized passenger seat **100** may include a leg assembly **103** that may mechanically couple or otherwise couple the generalized passenger seat **100** to a floor and/or the seat track of a passenger vehicle such as an airplane or other suitable passenger vehicle. As illustrated, the leg assembly **103** is a single-part leg assembly in which the leg assembly **103** contains one piece. Additionally, as illustrated, couplings associated with the leg assembly **103** may be fixed. While the leg assembly **103** is illustrated as a single component in FIG. 1, in other embodiments the leg assembly **103** may be a multi-part assembly.

(12) In certain embodiments, the generalized passenger seat **100** may include at least one tube **104**, and in certain embodiments, the generalized passenger seat **100** may include two or more tubes **104**. Accordingly, while two tubes **104a-b** are illustrated, the generalized passenger seat **100** may include any number of tubes **104** as desired, including a single tube **104** or more than two tubes **104**. The tubes **104** may include uniform tubes, non-uniform tubes, or a combination thereof. Moreover, the particular tubes **104** illustrated should not be considered limiting, and other portions of the seat frame component mounting systems **102** may include a tube **104** as desired. A fully assembled generalized passenger seat **100** may include various cushioning, a bottom seat pan, in-flight entertainment equipment, tray tables, and/or other components as desired. Optionally, the generalized passenger seat **100** may include a baggage bar **106** that may be coupled to the leg assembly **103**.

(13) According to certain embodiments of the present invention, and as best illustrated in FIGS. 2-5, the seat frame component mounting system **200** for a passenger seat may include a first tube mounting member **202**, a second tube mounting member **204**, and a fastener **206**. The seat frame component mounting system **200** may be the same as or similar to the seat frame component mounting system **102** discussed above with respect to FIG. 1. Each of the first tube mounting member **202**, the second tube mounting member **204**, the fastener **206**, and any other component of the seat frame component mounting system **200** may be formed from at least one of steel, steel alloy, stainless steel, titanium, aluminum, or any other material suitable for providing the necessary strength or other necessary physical and/or chemical characteristics of each of the components.

(14) In some embodiments, the first tube mounting member **202** may be integrally formed with a seat frame component **208**. Various seat frame components include a seat leg, a spreader, or any other suitable component of a seat frame. The second tube mounting member **204** may be separable from the first tube mounting member **202**. Additionally, the second tube mounting member **204** may be moveable relative to the first tube mounting member **202** along a linear axis, which may extend substantially along a central axis extending through the fastener **206** or may extend in any other suitable direction relative to the first tube mounting member **202**.

(15) As shown in FIGS. 2 and 5, the tube **104** may be positioned between the first tube mounting member **202** and the second tube mounting member **204**. The size of the first tube mounting member **202** and/or the second tube mounting member **204** may be interchangeable. For example, the diameter of each of the first tube mounting member **202** and the second tube mounting member **204** may be adjusted based on a size of the tube **104**.

(16) The fastener **206** may extend at least partially through the second tube mounting member **204**, the tube **104**, and the first tube mounting member **202** to couple the second tube mounting member **204** to the first tube mounting member **202** and secure the tube **104** between the second tube mounting member **204** and the first tube mounting member **202**. Examples of the fastener **206** may include but are not limited to screws, bolts, rivets or other mechanical or chemical fasteners. The fastener **206** may extend substantially through a center of an opening formed between the first tube mounting member **202** and the second tube mounting member **204**. However, the fastener **206** may also extend through the opening at any suitable angle and/or position.

(17) In some embodiments, the opening between the first tube mounting member **202** and the second tube mounting member **204** is sized so that the tube **104** is flush with a surface of the first

tube mounting member **202** and the second tube mounting member **204** when the tube **104** is positioned within the opening. With the tube **104** positioned within the opening between the first tube mounting member **202** and the second tube mounting member **204**, the fastener **206** may extend substantially through a center of the tube **104**. However, the fastener **206** may also extend through the tube at any suitable angle and/or position.

(18) In certain embodiments, an inner surface of at least one of the first tube mounting member **202** or the second tube mounting member **204** contours to an outer circumferential surface of the tube **104**. For example, with the fastener **206** coupling the first tube mounting member **202** to the second tube mounting member **204** with the tube **104** positioned between the first tube mounting member **202** to the second tube mounting member **204**, an entirety of an inner surface of at least one of the first tube mounting member **202** or the second tube mounting member **204** may be in contact with at least a portion of the outer circumferential surface of the tube **104**.

(19) At least one of the first tube mounting member **202** and the second tube mounting member **204** may provide a clamping force to the outer circumferential surface of the tube **104** via the fastener **206**. In some embodiments, the clamping force may be substantially evenly applied from at least one of the second tube mounting member **204** or the first tube mounting member **202** against the outer circumferential surface of the tube. Additionally, the clamping force may be applied to at least one third of the outer circumferential surface of the tube **104**, between one third to one half of the outer circumferential surface of the tube **104**, up to one half of the outer circumferential surface of the tube **104**, at least one half of the outer circumferential surface of the tube **104**, up to two thirds of the outer circumferential surface of the tube **104**, at least two thirds of the outer circumferential surface of the tube **104**, etc. Tightening and loosening the fastener **206** may adjust the amount of the clamping force applied to the tube **104**. The size of the first tube mounting member **202** or the second tube mounting member **204** may additionally be adjusted based on the clamping force that is to be applied against the tube **104**.

(20) In some embodiments, the fastener **206** may have a torque limitation based on a diameter of the fastener **206** and/or the material of the component being fastened, e.g., the first tube mounting member **202**, the second tube mounting member **204**, etc. Depending on the fastener **206** and materials of the first tube mounting member **202** and the second tube mounting member **204**, there may be a torque range for the fastener **206** based on the type of fastener used, the diameter of the fastener **206**, and the material of the portion of the component being fastened, e.g., the first tube mounting member **202**, the second tube mounting member **204**, etc. In an instance where the fastener **206** is over-tightened or over-torqued, any potential damage may be incurred by the fastener **206** rather than first tube mounting member **202**, the second tube mounting member **204**, or the tube **104**. Therefore, an adequate clamping force may be applied while preventing damage to the first tube mounting member **202**, the second tube mounting member **204**, or the tube **104**.

(21) As shown in FIGS. 3A-3B, at least one of the first tube mounting member **202** or the second tube mounting member **204** may include at least one guide member **304** and at least one opening **306**. Whichever of the first tube mounting member **202** or the second tube mounting member **204** includes the at least one guide member **304**, the other of the first tube mounting member **202** or the second tube mounting member **204** may include the at least one opening **306** so that each of the at least one guide members **304** has a corresponding opening **306** through which each of the at least one guide members **304** may extend through. The at least one guide member **304** may be press-fit bushings, pins, or any other suitable mechanical guide members.

(22) For example, as illustrated in FIGS. 3A-3B, the first tube mounting member **202** may include a first guide member **304a** and a second guide member **304b**, and the second tube mounting member **204** may include a first opening **306a** and a second opening **306b**. Each of the guide members **304a-b** and each of the openings **306a-b** may be positioned at equal distances from where the fastener **206** extends through the first tube mounting member **202** and/or the second tube mounting member **204**. Additionally, the second tube mounting member **204** may be positioned

relative to the first tube mounting member **202** so that the guide members **304a-b** substantially align with the openings **306a-b**. In fact, the guide members **304a-b** may assist in properly aligning the second tube mounting member **204** relative to the first tube mounting member **202**. Additionally, aligning the first tube mounting member **202** and the second tube mounting member **204** may involve aligning the first guide member **304a** with the first opening **306a** and the second guide member **304b** with the second opening **306b** so that the first guide member **304a** and the second guide member **304b** extend at least partially through the first tube mounting member **202** via the openings **306a-b**.

(23) In some embodiments, at least one second fastener **302** may be coupled to the first tube mounting member **202**. As illustrated in FIGS. **3A-3B**, the seat frame component mounting system **200** may include at least two second fasteners **302a-b**. Examples of the second fasteners **302a-b** may include but are not limited to screws, bolts, rivets, or other mechanical or chemical fasteners. The at least one second fastener **302** may extend at least partially through the at least one opening **306**, the at least one guide member **304**, and the first tube mounting member **202**. For example, the second fastener **302a** may extend at least partially through the first opening **306a**, the first guide member **304a**, and the first tube mounting member **202**, and the second fastener **302b** can extend at least partially through the second opening **306b**, the second guide member **304b**, and the first tube mounting member **202**. In some embodiments, the at least one second fastener **302** does not apply any additional clamping force from the second tube mounting member **204** or the first tube mounting member **202** to the tube **104**.

(24) As shown in FIG. **4**, at least one of the first tube mounting member **202** or the second tube mounting member **204** may include one or more first internal cavities **402a** and/or one or more second internal cavities **402b** connected to a respective one or more first internal cavities **402a**. For example, in the illustrated embodiment, the first tube mounting member **202** includes first internal cavities **402a** and second internal cavities **402b** connected to the first internal cavities **402a**. The first internal cavities **402a** and/or second internal cavities **402b** may be configured to receive one or more fasteners, such as second fasteners **302a-b**, and/or other components. Additionally, at least one of the first tube mounting member **202** or the second tube mounting member **204** may further include a third internal cavity **402c** that is configured to receive fastener **206** and/or other components. In some embodiments, one or more of the internal cavities may be at least partially threaded to facilitate coupling with the respective fasteners or other components. One or more of the internal cavities may also be sized to accommodate guide members, such as first guide member **304a** and/or second guide member **304b**, and/or other components. For example, the first internal cavity **402a** may have a first diameter (**D1**), and the second internal cavity **402b** may have a second diameter (**D2**). In some embodiments, the second diameter **D2** may be smaller than the first diameter **D1**. At least one cylindrical guide member (e.g., first guide member **304a** and/or second guide member **304b**) or other component may be positioned at least partially within the first internal cavity **402a**, but not within the second internal cavity **402b**. The at least one cylindrical guide member (e.g., first guide member **304a** and/or second guide member **304b**) may have a diameter greater than the second diameter **D2**, thereby preventing it from entering the second internal cavity **402b**. Furthermore, the cylindrical guide member (e.g., first guide member **304a** and/or second guide member **304b**) may include a hollow passageway that extends at least part of the length of the guide member, which may facilitate alignment or passage of components (e.g., second fasteners **302a-b**). The guide members **304a-b** may be friction-fit, snap-fit, or otherwise retained within the respective first internal cavity **402a**.

(25) As shown in FIG. **5**, with the first tube mounting member **202** coupled to the second tube mounting member **204** via the fastener **206**, there may be a gap **502** between the first tube mounting member **202** and the second tube mounting member **204**. The gap **502** may be in an area **504** of the first tube mounting member **202** and the second tube mounting member **204** that includes the at least one opening (e.g., openings **306** in FIG. **3**) and the at least one guide member

(e.g., guide members **304** in FIG. 3). As illustrated, there is a first gap **502a** between the first tube mounting member **202** and the second tube mounting member **204** in a first area **504a** of the first tube mounting member **202** and the second tube mounting member **204** that includes the first opening (e.g., first opening **306a** in FIG. 3) and the first guide member (e.g., first guide member **304a** in FIG. 3), and there is a second gap **502b** between the first tube mounting member **202** and the second tube mounting member **204** in a second area **504b** of the first tube mounting member **202** and the second tube mounting member **204** that includes the second opening (e.g., second opening **306b** in FIG. 3) and the second guide member (e.g., second guide member **304b** in FIG. 3). The area **504** of the first tube mounting member **202** and the second tube mounting member **204** that includes the at least one opening and the at least one guide member may be substantially planar.

(26) The gap **502** may permit minor movements of the second tube mounting member **204** relative to the first tube mounting member **202** due to various loads and/or torques applied to the passenger seat. In some embodiments, the minor movements may occur relative to and/or along the axes of the guide members **304**. By permitting such minor movements, the seat frame component mounting system **200** may further reduce undesired stresses and loads on the tube **104** thus reducing potential damage to the tube **104**.

(27) Referring now to FIG. 6, FIG. 6 shows an example method **600** for assembling a passenger seat having a tube and a seat frame component mounting system according to this disclosure. The example method **800** will be discussed with respect to the passenger seat **100** discussed relative to FIG. 1 and the seat frame component mounting system **200** discussed relative to FIGS. 2-5.

(28) At block **602**, the tube **104** is positioned between a first tube mounting member **202** and a second tube mounting member **204**. The second tube mounting member **204** may be separable from the first tube mounting member **202**.

(29) At block **604**, the fastener **206** is inserted at least partially through the second tube mounting member **204**, the tube **104**, and the first tube mounting member **202** to couple the second tube mounting member **204** to the first tube mounting member **202**.

(30) At block **606**, the fastener **206** is tightened to apply a clamping force from the second tube mounting member **204** and the first tube mounting member **202** substantially evenly against an outer circumferential surface of the tube **104**.

(31) At block **608**, the first tube mounting member **202** and the second tube mounting member **204** are aligned by aligning the at least one guide member **304** of the at least one of the first tube mounting member **202** or the second tube mounting member **204** with the at least one opening **306** of at least one of the first tube mounting member **202** or the second tube mounting member **204**.

(32) At block **610**, at least one second fastener **302** is coupled to the first tube mounting member **202**. The at least one second fastener **302** may extend at least partially through the at least one opening **306**, the at least one guide member **304**, and the first tube mounting member **202**.

(33) In the following, further examples are described to facilitate the understanding of the invention (and in some aspects, features of an apparatus or system described in one or more of these examples may be utilized in a method described in one of the other examples or vice versa). These examples are not meant to be mutually exclusive, exhaustive, or restrictive; and the disclosure is not limited to these examples but rather encompasses all possible modifications and variations within the scope of the issued claims and their equivalents.

(34) Example 1. A passenger seat (which may incorporate features of any of the subsequent examples) comprising: a tube; and a seat frame component mounting system comprising: a first tube mounting member; a second tube mounting member separable from the first tube mounting member, wherein the tube is positionable between the first tube mounting member and the second tube mounting member; and a fastener extending at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; and wherein a clamping force is

substantially evenly applied from the second tube mounting member and the first tube mounting member against an outer circumferential surface of the tube via the fastener.

(35) Example 2. The passenger seat of Example 1 or any of the preceding or subsequent examples, wherein the first tube mounting member is integrally formed with a seat frame component.

(36) Example 3. The passenger seat of Example 1 or any of the preceding or subsequent examples, wherein an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

(37) Example 4. The passenger seat of Example 1 or any of the preceding or subsequent examples, wherein at least one of the first tube mounting member or the second tube mounting member comprises at least one guide member.

(38) Example 5. The passenger seat of Example 4 or any of the preceding or subsequent examples, wherein the other of the at least one of the first tube mounting member or the second tube mounting member comprises at least one opening.

(39) Example 6. The passenger seat of Example 5 or any of the preceding or subsequent examples, wherein the second tube mounting member is positionable relative to the first tube mounting member so that the at least one guide member substantially aligns with the at least one opening.

(40) Example 7. The passenger seat of Example 6 or any of the preceding or subsequent examples, wherein at least one second fastener is coupleable to the first tube mounting member and extends at least partially through the at least one opening, the at least one guide member, and the first tube mounting member.

(41) Example 8. The passenger seat of Example 6 or any of the preceding or subsequent examples, wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.

(42) Example 9. A method (which may incorporate features of any of the preceding or subsequent examples) for assembling a passenger seat comprising a tube and a seat frame component mounting system, the method comprising: positioning the tube between a first tube mounting member and a second tube mounting member, wherein the second tube mounting member is separable from the first tube mounting member; inserting a fastener at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; and tightening the fastener to apply a clamping force from the second tube mounting member and the first tube mounting member substantially evenly against an outer circumferential surface of the tube.

(43) Example 10. The method of Example 9 or any of the preceding or subsequent examples, wherein an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

(44) Example 11. The method of Example 9 or any of the preceding or subsequent examples, wherein at least one of the first tube mounting member or the second tube mounting member comprises at least one guide member and at least one opening.

(45) Example 12. The method of Example 11 or any of the preceding or subsequent examples, further comprising aligning the first tube mounting member and the second tube mounting member by aligning the at least one guide member of the at least one of the first tube mounting member or the second tube mounting member with the at least one opening of at least one of the first tube mounting member or the second tube mounting member.

(46) Example 13. The method of Example 12 or any of the preceding or subsequent examples, further comprising coupling at least one second fastener to the first tube mounting member, wherein the at least one second fastener extends at least partially through the at least one opening, the at least one guide member, and the first tube mounting member.

- (47) Example 14. The method of Example 12 or any of the preceding or subsequent examples, wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.
- (48) Example 15. A seat frame component mounting system (which may incorporate features of any of the preceding or subsequent examples) comprising: a first tube mounting member; a second tube mounting member separable from the first tube mounting member and moveable relative to the first tube mounting member along a linear axis; and a fastener extending at least partially through the second tube mounting member and the first tube mounting member to couple the second tube mounting member to the first tube mounting member.
- (49) Example 16. The seat frame component mounting system of Example 15 or any of the preceding or subsequent examples, wherein the first tube mounting member is integrally formed with a seat frame component.
- (50) Example 17. The seat frame component mounting system of Example 15 or any of the preceding or subsequent examples, wherein the linear axis extends substantially along a central axis extending through the fastener.
- (51) Example 18. The seat frame component mounting system of Example 15 or any of the preceding or subsequent examples, wherein at least one of the first tube mounting member or the second tube mounting member comprises at least one guide member and at least one opening.
- (52) Example 19. The seat frame component mounting system of Example 18 or any of the preceding or subsequent examples, wherein the second tube mounting member is positionable relative to the first tube mounting member so that the at least one guide member substantially aligns with the at least one opening.
- (53) Example 20. The seat frame component mounting system of Example 19 or any of the preceding or subsequent examples, wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one guide member.
- (54) Different arrangements of the components depicted in the drawings or described above, as well as components and steps not shown or described are possible. Similarly, some features and sub-combinations are useful and may be employed without reference to other features and sub-combinations. Embodiments of the invention have been described for illustrative and not restrictive purposes, and alternative embodiments will become apparent to readers of this patent. Accordingly, the present invention is not limited to the embodiments described above or depicted in the drawings, and various embodiments and modifications may be made without departing from the scope of the claims below.

Claims

1. A passenger seat comprising: a tube; and a seat frame component mounting system comprising: a first tube mounting member; a second tube mounting member separable from the first tube mounting member, wherein the tube is positionable between the first tube mounting member and the second tube mounting member; and a fastener extending at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; wherein a clamping force is substantially evenly applied from the second tube mounting member and the first tube mounting member against an outer circumferential surface of the tube via the fastener; wherein at least one of the first tube mounting member or the second tube mounting member comprises a first internal cavity and a second internal cavity connected to the first internal cavity, the first internal cavity having a first diameter and the second internal cavity having a second diameter smaller than the first diameter,

and at least one cylindrical guide member positioned at least in part in the first internal cavity but not in the second internal cavity and having a diameter larger than the second diameter, the cylindrical guide member comprising a hollow passageway extending at least part of a length of the at least one cylindrical guide member; wherein the other of the at least one of the first tube mounting member or the second tube mounting member comprises at least one opening; wherein at least one second fastener is coupleable to the first tube mounting member, is separable from the at least one cylindrical guide member, and extends at least partially through the at least one opening, the hollow passageway of the at least one cylindrical guide member, and the first tube mounting member; wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one cylindrical guide member; and wherein the second tube mounting member is movable relative to the first tube mounting member while coupled to the first tube mounting member, and movement of the second tube mounting member either increases or decreases a size of the at least one gap.

2. The passenger seat of claim 1, wherein the first tube mounting member is integrally formed with a seat frame component.

3. The passenger seat of claim 1, wherein an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

4. The passenger seat of claim 1, wherein the second tube mounting member is positionable relative to the first tube mounting member so that the at least one cylindrical guide member substantially aligns with the at least one opening.

5. The passenger seat of claim 1, wherein an entirety of an inner surface of each of the first tube mounting member and the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

6. The passenger seat of claim 1, wherein the movement of the second tube mounting member occurs along and parallel to a longitudinal axis of the at least one cylindrical guide member.

7. A method for assembling a passenger seat comprising a tube and a seat frame component mounting system, the method comprising: positioning the tube between a first tube mounting member and a second tube mounting member, wherein the second tube mounting member is separable from the first tube mounting member; wherein at least one of the first tube mounting member or the second tube mounting member comprises a first internal cavity and a second internal cavity connected to the first internal cavity, the first internal cavity having a first diameter and the second internal cavity having a second diameter smaller than the first diameter, and at least one cylindrical guide member positioned at least in part in the first internal cavity but not in the second internal cavity and having a diameter larger than the second diameter, the cylindrical guide member comprising a hollow passageway extending at least part of a length of the at least one cylindrical guide member; and wherein the other of the at least one of the first tube mounting member or the second tube mounting member comprises at least one opening; inserting a fastener at least partially through the second tube mounting member, the tube, and the first tube mounting member to couple the second tube mounting member to the first tube mounting member; tightening the fastener to apply a clamping force from the second tube mounting member and the first tube mounting member substantially evenly against an outer circumferential surface of the tube; and coupling at least one second fastener to the first tube mounting member, wherein the at least one second fastener is separable from the at least one cylindrical guide member and extends at least partially through the at least one opening, the hollow passageway of the at least one cylindrical guide member, and the first tube mounting member; wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one cylindrical guide member; and wherein the second tube mounting member is

movable relative to the first tube mounting member while coupled to the first tube mounting member, and movement of the second tube mounting member either increases or decreases a size of the at least one gap.

8. The method of claim 7, wherein an entirety of an inner surface of at least one of the first tube mounting member or the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

9. The method of claim 7, further comprising aligning the first tube mounting member and the second tube mounting member by aligning the at least one cylindrical guide member of the at least one of the first tube mounting member or the second tube mounting member with the at least one opening of at least one of the first tube mounting member or the second tube mounting member.

10. The method of claim 7, wherein an entirety of an inner surface of each of the first tube mounting member and the second tube mounting member is in contact with at least a portion of the outer circumferential surface of the tube.

11. The method of claim 7, wherein the movement of the second tube mounting member occurs along and parallel to a longitudinal axis of the at least one cylindrical guide member.

12. A seat frame component mounting system comprising: a first tube mounting member comprising an arcuate portion; a second tube mounting member comprising an arcuate portion, the second tube mounting member being separable from the first tube mounting member; and a fastener extending at least partially through the arcuate portion of the second tube mounting member and the arcuate portion of the first tube mounting member to couple the second tube mounting member to the first tube mounting member; wherein the second tube mounting member is movable relative to the first tube mounting member along and parallel to a linear axis while the second tube mounting member is coupled to the first tube mounting member; wherein at least one of the first tube mounting member or the second tube mounting member comprises a first internal cavity and a second internal cavity connected to the first internal cavity, the first internal cavity having a first diameter and the second internal cavity having a second diameter smaller than the first diameter, and at least one cylindrical guide member positionable in the first internal cavity but not in the second internal cavity and having a diameter larger than the second diameter, the cylindrical guide member comprising a hollow passageway extending at least part of a length of the at least one cylindrical guide member; wherein the other of the at least one of the first tube mounting member or the second tube mounting member comprises at least one opening; and wherein at least one second fastener is coupleable to the first tube mounting member, is separable from the at least one cylindrical guide member, and extends at least partially through the at least one opening, the hollow passageway of the at least one cylindrical guide member, and the first tube mounting member into the second internal cavity.

13. The seat frame component mounting system of claim 12, wherein the first tube mounting member is integrally formed with a seat frame component.

14. The seat frame component mounting system of claim 12, wherein the linear axis extends substantially along a central axis extending through the fastener.

15. The seat frame component mounting system of claim 12, wherein the second tube mounting member is positionable relative to the first tube mounting member so that the at least one cylindrical guide member substantially aligns with the at least one opening.

16. The seat frame component mounting system of claim 15, wherein there is at least one gap between the first tube mounting member and the second tube mounting member in an area of the first tube mounting member and the second tube mounting member comprising the at least one opening and the at least one cylindrical guide member.

17. The seat frame component mounting system of claim 16, wherein movement of the second tube mounting member either increases or decreases a size of the gap, and the movement of the second tube mounting member occurs along and parallel to a longitudinal axis of the at least one cylindrical guide member.

